Agamjot Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.73
The Cambridge Int. School, Kudlu, Bengaluru, Karnataka, India	Class XII (CBSE)	2024	97.60%
Vibgyor High School, Bengaluru	Class X (CISCE)	2022	97.40%

SCHOLASTIC ACHIEVEMENTS

Received A+ (outstanding performance) in the following courses: Computer architecture, Probability, Circuits and Network analysis, Signal processing, Control systems

PROJECTS

RISCV 64-bit RV64IZicsr core with xv6 kernel (Github)

- 5 stage pipeline running at 30MHz on the **PYNQ Z2 FPGA.**
- Compliance tested with official **riscv-tests** (rv64i) suite with **100% pass rate.**
- Benchmarked with **Dhrystone** with ~13k Dhrystones/sec (0.25 DMIPS/MHz) performance.
- Automatic **verification pipeline** with register and memory hexdump matching with **Spike simulator.**
- **MMU** built for memory virtualization compatible with **SV39 3-level pagetables**.
- **3 privilege levels** with appropriate **trap** (interrupt/exception) mechanisms and delegations.
- **L1 64-entry direct mapped TLB** and **L1 direct mapped 128KiB split I and D cache** for efficient memory operations.
- **AXI4 Burst** memory controller for interaction with DDR3 memory through AXI High performance Slave ports in ZYNQ PS.
- **Font rendering VGA engine** for displaying 160x45 terminal interface on external 720p character display.
- **Correlating branch predictor** with Local History and Predictor tables giving **~17% increase in IPC** on dhrystone benchmark.
- **HPM** and cycle counters (partial **Zicntr** and **Zihpm** extension).
- Built **xv6 drivers** for custom disk interface to read/write SD Card along with **AXI4 Lite** hardware controllers.

Tinkering with xv6 (Github)

- Added **lazy page allocation** and **COW forks** for efficient memory management.
- Extended filesystem to support **double indirect inodes** for bigger files.
- Efficient memory allocator with **per-CPU freelists** and **page stealing** to reduce lock contentions.
- Buffer cache enhancements with **per-bucket locks** on the block hash table to reduce lock contention on concurrent lookups.
- **Symbolic link** functionality with link resolution logic in the file-open path and **O_NOFOLLOW** support.
- Implemented unix-style **periodic signal delivery** (SIGALARM equivalent) with user-level context switch.

R2SDF Pipelined FFT hardware accelerator (Github)

- Implemented **Decimation in Frequency** (DIF) 8-point FFT accelerator in Verilog.
- Streaming **Radix-2 Single Delay Feedback** design with 18-bit (Q6.12) fixed-point format with streaming data.
- Pipelined **butterfly** and **decimator stages** along with a **dual reorder buffer** for max throughput.
- Performed synthesis on Intel Quartus Lite with Fmax of **35MHz** and **204 ALMs & 400 Regs** usage.

MINI PROJECTS

Eigenvalue computation of complex matrices (Github)

- Efficient algorithm for finding eigenvalues of matrices with complex entries in C.
- **Householder transformation** techniques for reduction of matrix to Hessenberg form.
- **Givens rotation** for repeated QR decomposition of the hessenberg matrix.
- Improved QR convergence rates with **Rayleigh Shifts** between iterations.

Inter IIT Tech Meet 14.0 Drona aviation Low Prep PS (IIT Patna)

- Contributed to achieving **on-board stabilization** of a GPS-denied nano-drone.
- Improved **driver code** for optimal low light **(~15 LUX)** function of **Optical Flow Sensor (PAW3903).**

Hardware approximate division unit (Github)

- Implemented a multi-cycle parameterizable approximate division unit using **Newton-Raphson** method.
- **Optimal initial-guess generation** using LUT based piecewise approximation.
- **Iterative refinement** along with **normalization denormalization** to achieve minimal error.

RELEVANT COURSES

Computer Architecture (CS2323), Data structures and applications (ID2230), Digital Systems Theory & Lab (EE1201 & EE1501), Probability (EE1080), Signal processing (EE2000), Semiconductor Device fundamentals (EE2103)

SKILLS

Languages: C, Verilog, Python, Makefile, RISCV Assembly

Tools and EDA: Vivado, LTSpice, Git, GDB, Verilator, Altium, Vim

POSITIONS OF RESPONSIBILITY

Core member of Robotix Club, IITH

Core member at Elektronika, IITH (Electronics and Signal Processing Club):

Collaborating on a hardware graphics accelerators project in a 5-member team.

Teaching assistant for EE1080 Probability:

Conducted sessions for clarifying concepts and performed programming assignment grading.

TEDxIITHyderabad PR and Networking Coordinator